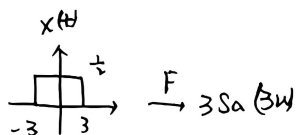


# 第6次作业

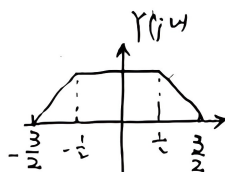
3-19 3-20 3-23 (1)(2) 3-24 (1)(2)(4) 3-26 27 (1)(3) 33 34(1)



$$\cos W = \frac{e^{jW} + e^{-jW}}{2}$$

$$\frac{x(t-1) + x(t+1)}{2} \xrightarrow{F} 3Sa(3W) \cos W = H(jW)$$

$$h(t) = \begin{cases} \frac{1}{4} & -2 < t \leq 4 \\ \frac{1}{2} & t \leq 2 \\ 0 & \text{其它} \end{cases}$$



$$p(t) = \frac{1}{\pi} \sum_{k=-\infty}^{\infty} e^{jk\pi t} \quad x(t) p(t) \xrightarrow{F} \frac{1}{\pi} \sum_{k=-\infty}^{\infty} X(jk\pi)$$

$$p(t) = \frac{1}{4\pi} \sum_{k=-\infty}^{\infty} e^{jk\pi t} \quad x(t) p(t) \xrightarrow{F} \frac{1}{4\pi} \sum_{k=-\infty}^{\infty} X(jk\pi)$$

$$y(t) = \frac{2E}{\pi} \left\{ \text{Si}[W_c(t-t_0)] - 2\text{Si}[W_c(t-t_0-T)] + 2\text{Si}[W_c(t-t_0-2T)] - \text{Si}[W_c(t-t_0-3T)] \right\}$$

3-20

$$X(jW) = \frac{Y(jW)}{H(jW)} = \frac{1}{jW+4}$$

$$x(t) = e^{-4t} u(t)$$

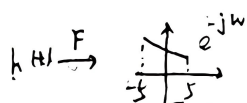
3-26

$$x(t) = \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$y(t) = \sum_{k=-\infty}^{\infty} a_k H(jk\omega_0) e^{jk\omega_0 t}$$

频率分量只可能减少 ( $H(jk\omega_0) = 0$ )  
不可能增多 且仍是周期信号

3-27 (1)

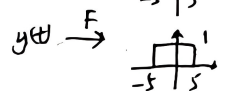
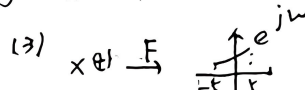


$$x(t) \xrightarrow{F} \frac{1}{\pi} \int_{-5}^5 \left[ S\left(\frac{W}{2} - 1\right) + S\left(\frac{W}{2} + 1\right) \right]$$

$$x(t) * h(t) \xrightarrow{F} 0$$

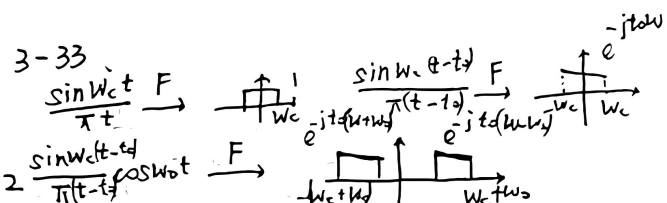
$$y(t) = x(t) * h(t) = 0$$

3-28



$$y(t) = \frac{\sin \frac{\pi t}{2}}{\pi t}$$

3-33



$$h(t) = 2 \frac{\sin W_c(t-t_0)}{\pi(t-t_0)} \cos W_0 t$$

非因果 物理上不可实现

3-29

(1)

$$y(t) = x(t) \sum_{k=-\infty}^{\infty} a_k e^{jk\omega_0 t}$$

$$Y(jW) = X(jW) \sum_{k=-\infty}^{\infty} a_k \delta(W - k\omega_0)$$

(2)

$$p(t) = \cos \frac{\pi t}{2} = \frac{1}{2} (e^{j\frac{\pi t}{2}} + e^{-j\frac{\pi t}{2}})$$

$$Y(jW) = \frac{1}{2} [X(j(W - \frac{\pi}{2})) + X(j(W + \frac{\pi}{2}))]$$